Connectivity and User Interface

Shared monitoring, target adjustment and experiment logging

AT THE REACTOR | **NETWORK SERVICE** | **ON THE OPERATOR LAPTOP**

Control board
Sensors + actuators
Local feedback loops

UART ⟷

ESP32 gateway
Forwards readings
and target requests

Wi-Fi MQTT ⟷

MQTT broker
Routes messages
between clients

MQTT ⟷

Laptop backend
Hosts UI + CSV logs
Feeds data analysis

Local web ⟷

Browser
Dashboard
Operator inputs

Readings + controller confirmations → ← Target requests

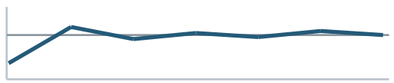
PROPOSED SCREEN / example values

Temperature
29.8 °C
Confirmed: 30.0 °C
New target | Apply



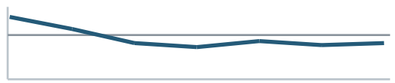
25–35 °C

Stirring speed
998 RPM
Confirmed: 1000 RPM
New target | Apply



500–1500 RPM

pH
5.02
Confirmed: 5.00
New target | Apply



pH 3–7

OPERATOR ACTIONS

Monitor values and trends.

Apply validated setpoints.

Wait for controller confirmation.

Export timestamped records.

On link loss: mark readings stale.
Local loops follow the agreed policy.

PROPOSED TESTS 1 Hz reporting | 95% of confirmations ≤2 s | stale warning after 5 s

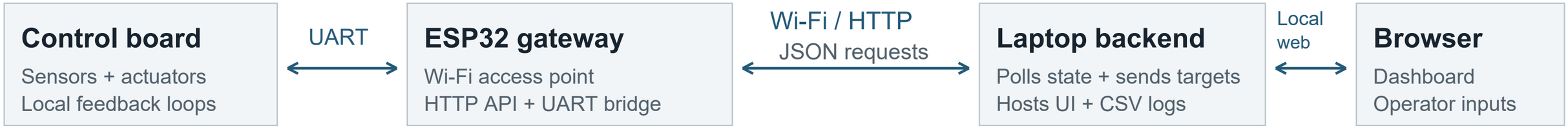
Proposed architecture. Verify board models and course-broker access. Brief: Design Specification pp. 4, 13, 17–18; Connectivity pp. 18–19.

Connectivity and User Interface

Alternative proposal: direct HTTP over a local Wi-Fi connection

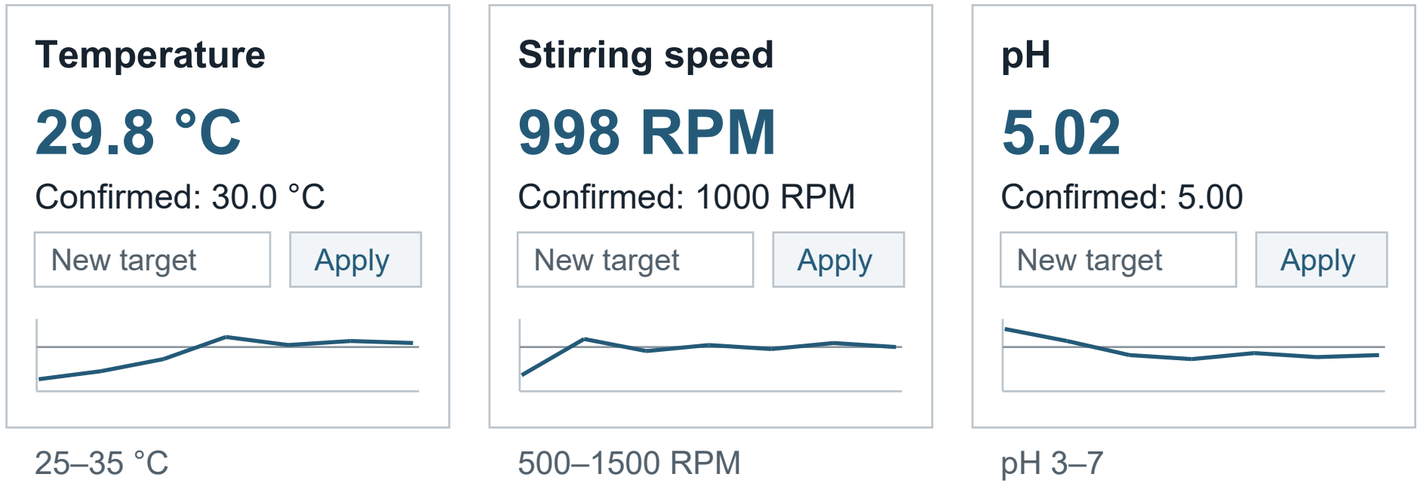
AT THE REACTOR

ON THE OPERATOR LAPTOP



Laptop joins the ESP32 network. Backend polls readings and sends target requests.

PROPOSED SCREEN / example values



25–35 °C | 500–1500 RPM | pH 3–7

OPERATOR ACTIONS

Monitor values and trends.

Apply validated setpoints.

Wait for controller confirmation.

Export timestamped records.

On link loss: mark readings stale.

Local loops follow the agreed policy.

PROPOSED TESTS 1 Hz reporting | 95% of confirmations ≤2 s | stale warning after 5 s

Local access only: confirm this scope and access-point use with staff. Internet access is not provided by this arrangement.

Backup: command flow and software structure

How a user request becomes a confirmed controller target

COMMAND FLOW

User request
Edit target + Apply

→

Validate request
Range + message ID

→

Send to controller
Network, then UART

→

Controller checks
Accept or reject

→

Return result
Update UI + event log

Invalid → show error; do not send.

No result → show timeout / unconfirmed.

An applied target is confirmed by the control board. Reaching that target takes longer.

SOFTWARE STRUCTURE

Gateway firmware

- UART messages
- Network transport
- Link status

Laptop application

- UI + commands
- Logging + analysis
- Device connection

First integration test: one simulated reading → dashboard, then one target request → controller confirmation.

Supporting models for Design Specification p. 18. Proposed responsibilities and logic; not measured implementation results.